

— FROZEN ENCODERS · AUTORESEARCH · SEARCH

Test-Time Compute for Information Retrieval

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SCAN FOR THE SLIDES



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Test-time compute is for large reasoning models. Small models, the story goes, have nothing to gain.

- | BEST-OF-N Sample many candidates, keep the best.
- | SELF-CONSISTENCY Majority-vote over sampled chains of thought.
- | VERIFIER SEARCH Spend inference FLOPs, climb a smooth Pareto curve.

A smaller model spends inference to match a bigger one. To our knowledge, dense retrieval, dominated by small embedding models, has not been studied through this lens.

Search is test-time compute.

You wire trained embeddings, rerankers, multi-vector retrievers, and query expanders into a **pipeline at inference** to squeeze out relevance. A small model does not have to get bigger. It has to assemble more search at test time.

DON'T SCALE IT

one keyword match. Cheap, and probably not good enough.

SCALE IT

embed, retrieve, rerank, expand, fuse. More inference can buy more relevance.

The question is not whether the model is big enough. It is how much search pipeline you assemble at test time, and whether it pays off.

Two ways to manufacture that pipeline at test time.

A RECOMBINE THE GEOMETRY

An agentic loop writes **programs** over one frozen encoder: chunk, z-score, fuse channels, feed back. The pipeline is multi-pass embedding algebra.

[the paper](#) · [this talk's core](#)

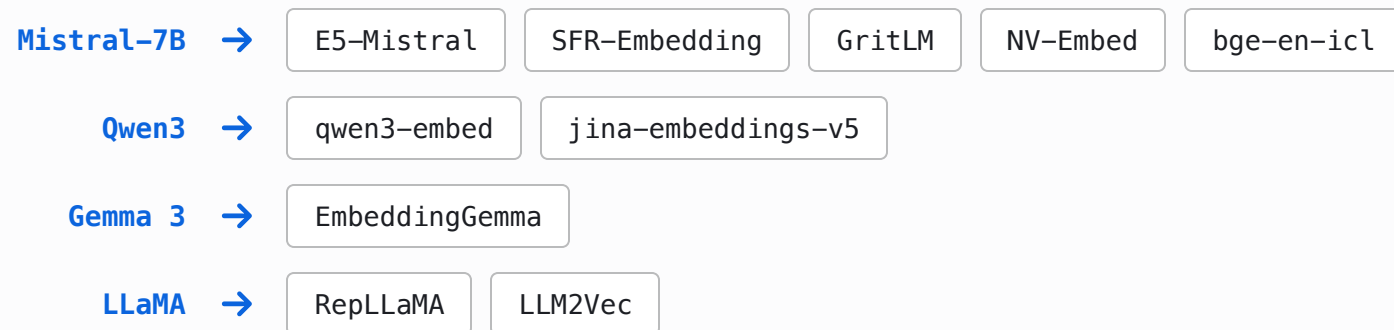
B COMPOSE THE TOOLS

A small agent LLM **wires retrieval tools** (grep, embed, rerank, select_diverse) over a corpus under a budget. The pipeline is an agentic tool chain.

[searchbox](#) · [later](#)

Same move at two altitudes:
[assemble structure at inference, not a bigger model.](#)

But today's small embedders are distilled or adapted from LLM backbones.



The lineage from a large language model to a small deployable encoder is now the dominant recipe, not the exception.

If test-time compute lives in the LLM representation space, these distilled encoders should inherit it. [Do they?](#)

Scoring runs from one cosine to late interaction.

SINGLE-VECTOR COSINE

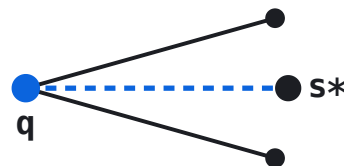


$$\cos(q, d)$$

one vector per document

FROZEN BASELINE

SENTENCE MAXSIM

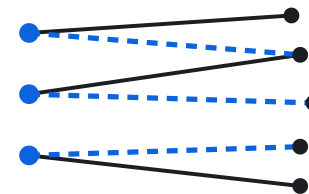


$$\max_s \cos(q, s)$$

same encoder, document split into
sentence vectors

TEST-TIME STRUCTURE

COLBERT MULTI-VECTOR



$$\sum_i \max_j \cos(q_i, d_j)$$

every query token, max over all doc
tokens

NEEDS A MULTI-VECTOR MODEL

Test-time compute adds **structure over the same frozen vectors**: chunk, feed back, fuse, and climb toward late interaction with no new model.

How much can a frozen single-vector encoder gain at inference alone?

NO RETRAINING

NO AUXILIARY MODEL

NO LEARNED PARAMETERS

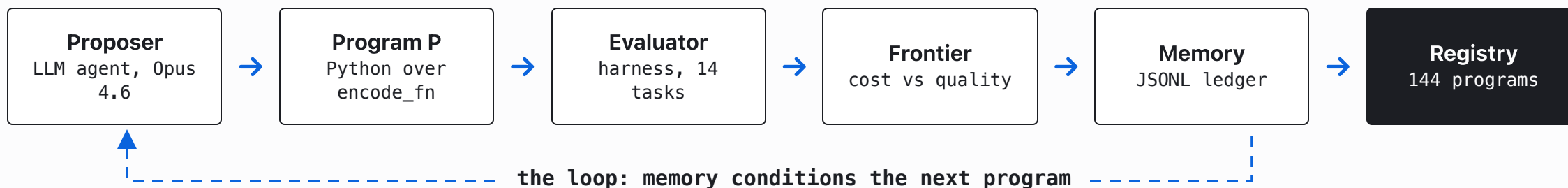
ONE FROZEN ENCODER API

Prior test-time methods for retrieval each break at least one rule:

- | HYDE / QUERY2DOC an external LLM in the query path.
- | GQR a second, supervisory retriever.
- | METAEMBED trained extra parameters at deployment.

We forbid all three, and ask whether the gain scales with the compute spent.

An LLM agent writes programs over the frozen encoder, and a harness scores them.



$P : (\mathbf{Q}, \mathbf{D}, \mathbf{S}, \mathbf{ctx}) \rightarrow \mathbf{S}'$ an arbitrary deterministic Python function over the baseline cosine similarity. Each encode_fn call is one unit of test-time compute.

Each generation: propose a program, score it on 14 tasks, remember everything.

THE PROGRAM INTERFACE

```
rerank(q_emb, d_emb, sim, *,
      encode_fn, q_texts, d_texts) -> scores
```

encode_fn is the test-time-compute budget. It re-encodes any text, switches the LoRA adapter (retrieval / passage / classification / text-matching), picks a Matryoshka dimension, or caps length. One call is one unit of compute.

THE MEMORY

A JSONL ledger, one row per program: per-task Δ nDCG@10, win/tie/loss, cost ratio, parent, a novelty claim, and a post-hoc lesson. The proposer reads the frontier source plus this history, so each round builds on the last.

SIX TABOOS, AUTO-REJECT

- ✗ **No hyperparameters:** only structural choices (argmax, mean, median, the 0.5 midpoint)
- ✗ **No stochastic elements:** deterministic given the input
- ✗ **No task-specific routing:** one program, all 14 tasks
- ✗ **No external models:** only the frozen encoder
- ✗ **No learned parameters**
- ✗ **No trivial variants:** differ in operation, not in constants

The taboos force the search toward universal structure, not a per-task tuned config.

Two things a program can spend: compute, or structure.

TEST-TIME COMPUTE

The extra forward passes the program runs through the encoder.

$$c = \frac{T_{\text{base}} + T_{\text{prog}}}{T_{\text{base}}}$$

Spend more inference.

TEST-TIME STRUCTURE

A fixed recombination of the frozen geometry: chunking, z-scoring, channel fusion, centroid feedback.

deep structure at $c \approx 1$

Almost no extra compute.

The two axes are independent. The whole question: **which one converts into quality that transfers?**

We run the search under two admission rules.

COMPUTE-SPENDING

Admit a program if its in-domain mean $\Delta n\text{DCG}@10$ beats every prior one. The proposer is encouraged to spend more inference.

→ a frontier from $c = 1.2$ to 14.7

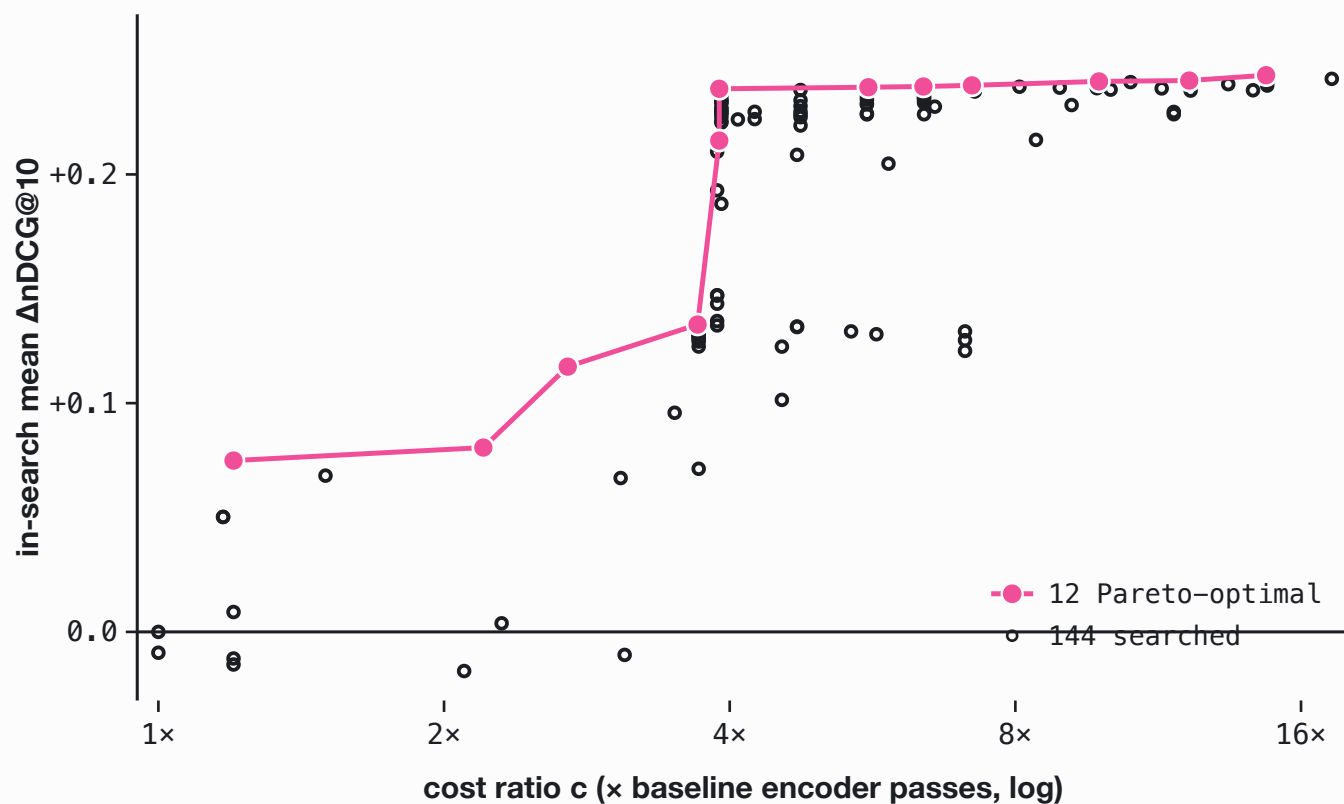
TRANSFER-SELECTING

Admit only if a held-out validation split improves with no task regressing past 0.05. No reward for spending compute.

→ converges near $c = 1$

Neither search ever sees the 19 held-out evaluation tasks or the unseen encoder families.

Told to spend compute, the search draws a clean Pareto curve.



144

programs searched over 144 generations

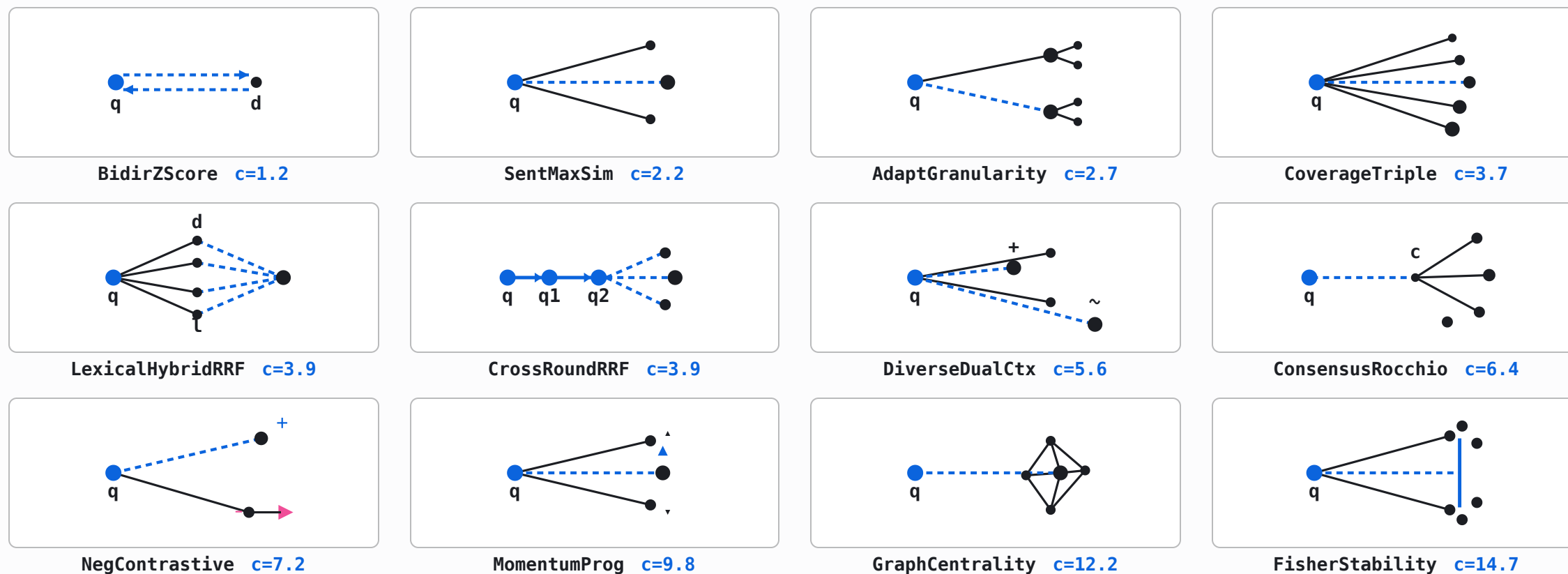
12

Pareto-optimal, from $c = 1.2$ to 14.7

In-search mean $\Delta nDCG@10$ climbs from +0.07 to +0.24. This looks exactly like test-time-compute scaling.

In-search only. The held-out test comes next.

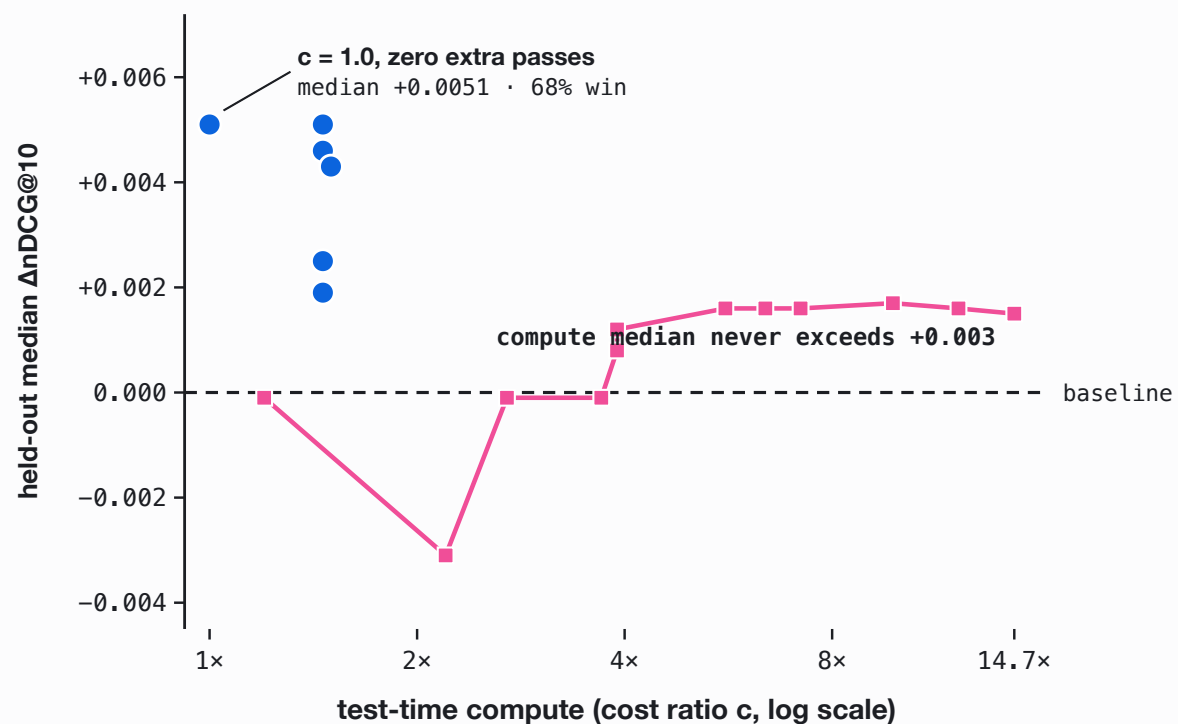
Twelve programs on the frontier, one frozen encoder.



Every one is a training-free recombination of the same frozen vectors: chunking, z-scoring, feedback, fusion. Cost climbs left to right; the gains, as the next slide shows, do not.

On unseen encoders, compute is flat. Cheap structure is not.

■ compute-spending ■ transfer-selecting (structure)



14.7×

the most compute a program spends, beaten flat by a zero-pass program at $c = 1.0$

-0.016

compute's pooled held-out mean, below baseline

-0.98

its worst per-query cell

The transfer-selecting program at $c = 1.0$, zero extra forward passes, already beats the most expensive compute program at $c = 14.7$. Median is plotted, robust to the -0.98 tail; the pooled mean is negative too.

Transferable programs cost at most 1.5x. The best costs nothing.

PROGRAM	<i>c</i>	MEDIAN	WIN-RATE	MEAN	WORST CELL
ZeroCost-Consensus	1.00	+0.0051	68.4%	+0.0148	-0.1070
Deca-Axis	1.46	+0.0051	68.4%	+0.0133	-0.0365
Dodeca-v077	1.46	+0.0046	66.7%	+0.0126	-0.0420
Transpose-Consensus	1.50	+0.0043	83.3%	+0.0219	0.0000
Cross-Axis	1.46	+0.0025	64.9%	+0.0097	-0.0240
Penta-Axis	1.46	+0.0019	59.6%	+0.0107	-0.0485

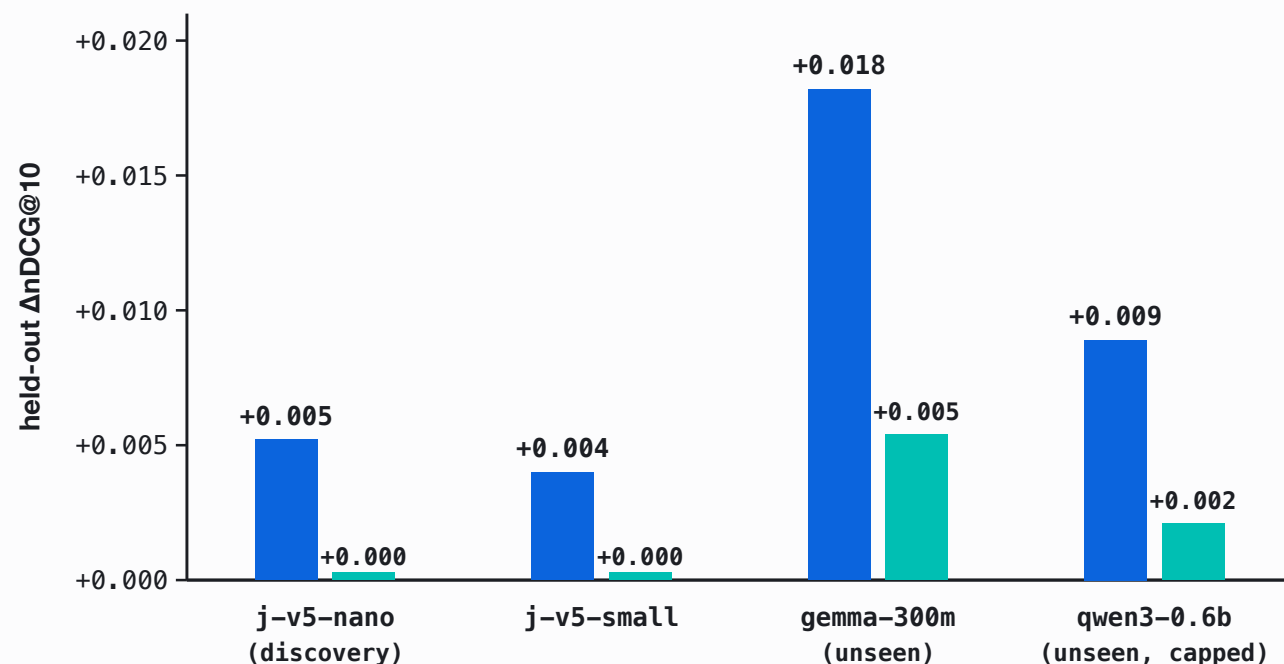


losing cells: Transpose-Consensus, at an 83% held-out win-rate

Held-out quality is negatively correlated with cost ($r = -0.37$). The worst cell over the whole table is -0.107: nine times smaller than the compute frontier's -0.98 collapse. The transfer frontier turns the held-out mean positive by removing the catastrophic tail, not by relying on a higher win-rate.

Gains are largest on encoder families never seen in discovery.

mean median



Discovered on j-v5-nano. Mean $\Delta nDCG@10$ is positive on all four families, and largest on the two never seen during discovery. The transfer follows general embedding geometry, not artifacts of the discovery encoder.

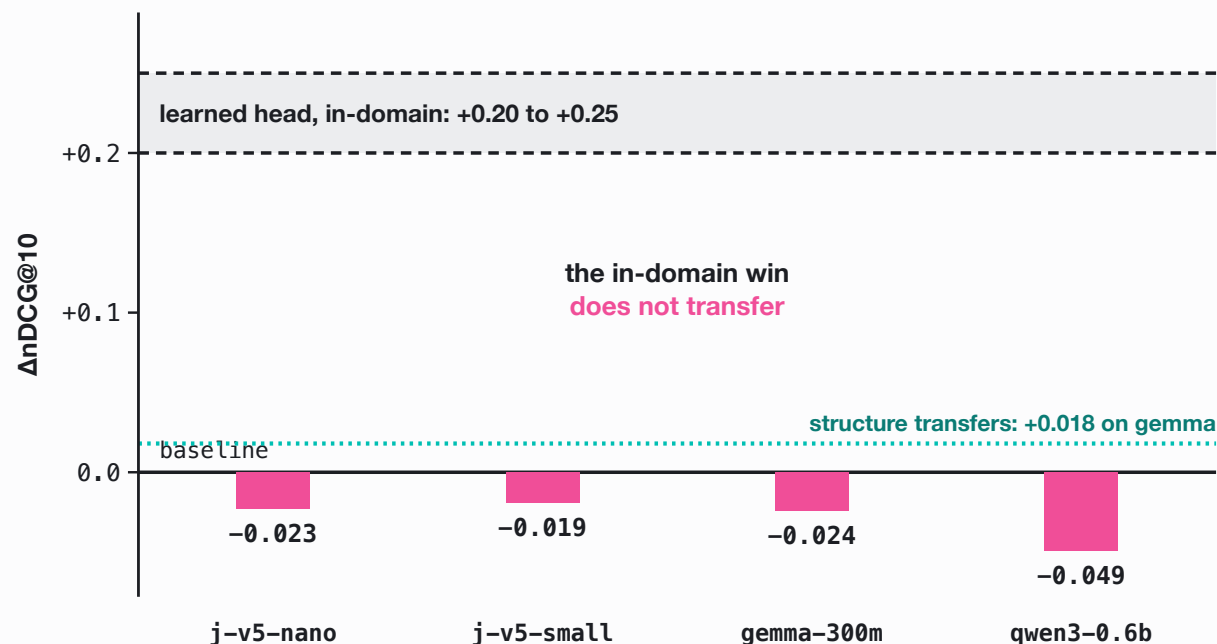
ACROSS LANGUAGES, T00

Applied unmodified to French and Greek: median **+0.016**, an **86%** win-rate, and every held-out cell positive on gemma-300m.

— WHY NOT JUST TRAIN A HEAD?

A matched-budget learned head memorizes. Structure transfers.

□ in-domain ■ held-out ■ structure (ref)



Trained on the same 14 discovery tasks, a linear, low-rank, or MLP head improves in-domain retrieval by **+0.20 to +0.25** $\Delta nDCG@10$, yet falls below baseline on **every** held-out encoder.

Adding parameters at the same data budget does not transfer. Recombining the frozen geometry does.

The transferable programs are classical IR, re-derived in embedding space.

Reciprocal Rank Fusion

REDISCOVERED, NOT SEEDED

Fisher Linear Discriminant

REDISCOVERED, NOT SEEDED

Rocchio Pseudo-Relevance Feedback

OPERATIONALIZED FROM A SEEDED IDEA

Sentence-level MaxSim

OPERATIONALIZED FROM A SEEDED IDEA

The reusable structure is geometric and old: z-scoring, sub-document granularity, centroid feedback. It depends on cosine geometry, not on model-specific training, which is why it carries to encoder families never seen during discovery.

The catch: the loop optimizes exactly what you tell it.

V2 OPTIMIZED WHAT WE TOLD IT

- fake** in-domain mean, plus "spend more compute", gave a fake scaling frontier to $c = 14.7$ that already saturated by $4\times$
- 0.98** a **tail-blind** objective, so held-out cells collapse (gemma 0.99 to 0.01)
- all 12** held-out mean negative for **every** program: transfer was never in the loss

V3 OPTIMIZES THE CLAIM

- split** train / a **hidden validation gate** / test touched once: select on validation, report on test
- floor** validation median with a **worst-case floor**: kills the catastrophic tail at the source
- strip** **paired-bootstrap** admission, and **drop the compute mandate** from the prompt

A loop hill-climbs the metric you wrote, not the one you meant.

Anti-patterns the ledger ruled out: pseudo-doc-as-query -0.49, stacking

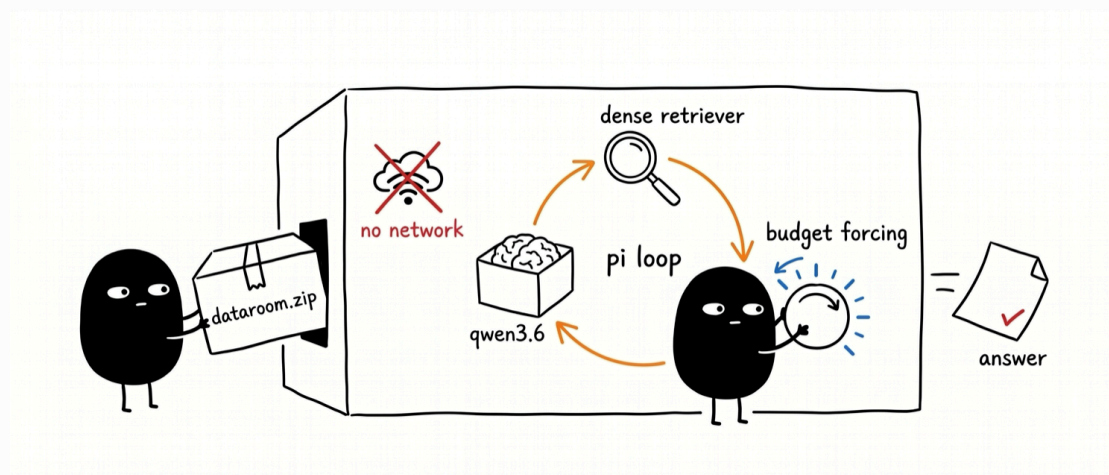
four LoRAs -0.45, first-100-chars truncation -0.43.

searchbox: a small agent wires the pipeline itself.

A self-hosted Qwen3.6-35B-A3B (~3B active) in an **airgapped** harness explores a .zip dataroom under a budget, composing its own pipeline from local tools: grep, embed, rerank, similarity, cluster, select_diverse. No network, near-zero token cost. **EXPERIMENTAL**

OPEN RESEARCH QUESTIONS

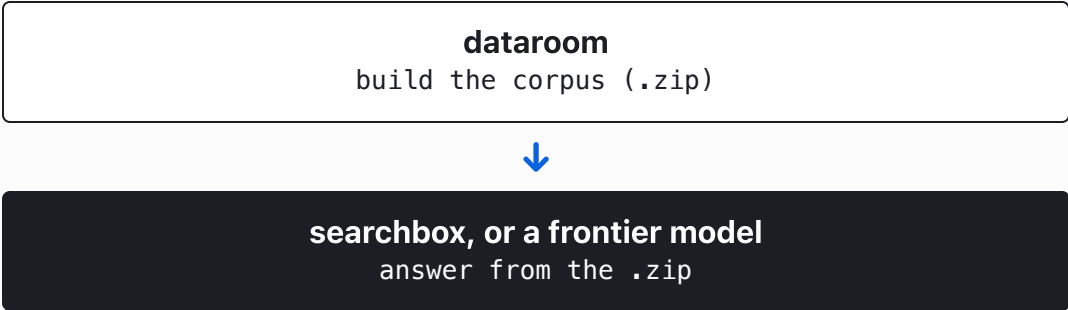
- Which tool does the agent reach for first?
- Is grep all you need: where does a dense retriever add nothing?
- Does forcing more token budget (scaling TTC) help on the hard questions?



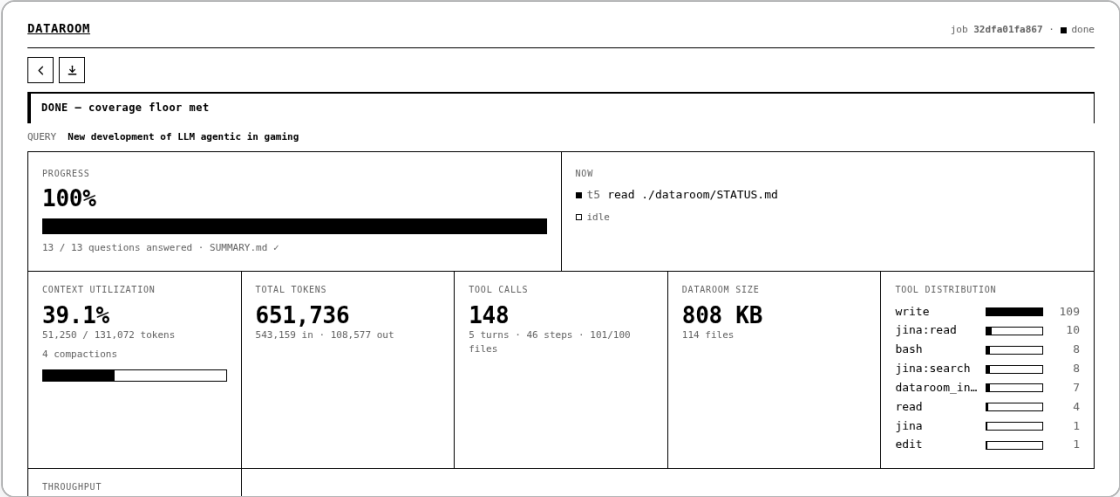
the airgapped loop, illustrated · live at hanxiao.io/searchbox

dataroom: a small model builds the corpus first.

Same harness, one stage earlier. A local model loops **search-read-write** into a fully cited .zip corpus, built for a machine to consume rather than a human to skim. Research is mechanical, so it runs on your own GPU at near-zero token cost until the dataroom is comprehensive.



Stage one of two: hand the grounded .zip to searchbox or a frontier model for the expensive second stage.



Live job page: progress to the coverage floor, token usage, and the tool-call distribution. Outcome-based stopping, not a token budget.

Both manufacture a search pipeline at test time. Neither grows the model.

	A EMBEDDING PROGRAMS	B SEARCHBOX
who assembles it	an agentic search loop, offline	a small agent LLM, at query time
the pipeline is	multi-pass embedding algebra	a chain of retrieval tools
the small model	frozen encoder, 239M	Qwen3.6, ~3B active
what scales	structure, not forward passes	tool composition, not parameters

Test-time compute for small models is **pipeline assembly**, not raw compute.

For frozen retrieval, test-time structure substitutes for model size. Test-time compute does not.

- 01 Small distilled models have real, transferable test-time potential, realized as structure, not raw compute.
- 02 Two ways to spend it: recombine the encoder's geometry, or let an agent compose the tools. Both assemble a pipeline at inference.
- 03 Trust the gain only when the loop optimizes transfer, with a held-out gate it cannot see and a worst-case floor.

~1x

query-time cost. 0.011 s of algebra against 40 s of encoding: the axis that transfers is also the cheapest to serve.

Honest caveat: absolute held-out gains are modest, pooled means of +0.004 to +0.018 Δ nDCG@10. The claim is about the shape of the compute-quality relation and the tail, not magnitude.

— THANK YOU

Spend structure, not compute.

Zero training. Across frozen encoders. About 1× serving cost.

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`github.com/hanxiao / embedding-ttc · searchbox · dataroom`

`arXiv:2605.11374`

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THESE SLIDES



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